

## Agent Design Spec: Dataset Scout

**1. Job to be Done** The agent will act as a specialized computer vision research scout. Its sole job is to search, filter, and summarize highly relevant datasets exclusively from Kaggle and Roboflow Universe based on user-provided keywords, outputting verified download links, dataset sizes, class lists, and supported export formats (e.g., YOLO).

**2. User and Usage Frequency** **User:** Me (Software Engineering student and AI developer). **Frequency:** 1-2 times a week, typically during the initial research phase of a new machine learning project or when restructuring an existing dataset for version iterations.

### 3. Tools and Data Needed & Access Plan

- **Tool:** Native Web Search / Browsing capability.
- **Data Source:** Public web pages on [kaggle.com](https://www.kaggle.com) and [universe.roboflow.com](https://universe.roboflow.com).
- **Access Plan:** I will use the built-in search tool provided by the Claude platform (Claude Pro/Project). Because it uses native web browsing to query public repositories, no external API keys, authentication tokens, or complex scraper setups are required. The access plan is immediate and fully covered by the platform.

### 4. Draft Instructions (System Prompt)

"You are an expert Machine Learning Data Scout. Your job is to find computer vision datasets based on the user's keywords. You must strictly use your web search tool to query [site:kaggle.com](https://www.kaggle.com) and [site:universe.roboflow.com](https://universe.roboflow.com). For every dataset you find, you must extract and present: 1) The exact, working URL, 2) The number of images/instances, 3) The specific classes available, and 4) Supported formats (e.g., YOLO, COCO). You must verify that the link is active before providing it. If you cannot find a highly specific dataset, do not guess; suggest the closest available alternatives."

### 5. Pre-Build Eval Cases (FL-03 Style)

- **Eval Case 1:**
  - *Input:* "Find a YOLO format dataset on Roboflow for detecting ground obstacles and potholes."
  - *Expected Output:* Returns 2-3 verified Roboflow Universe links containing pothole/road anomaly datasets, explicitly listing the image counts and confirming YOLOv8 export availability.
- **Eval Case 2:**
  - *Input:* "Search Kaggle for an aerial view/UAV visual odometry dataset."
  - *Expected Output:* Returns verified Kaggle links (like drone-view object detection or semantic segmentation datasets), filtering out ground-level images.
- **Eval Case 3:**
  - *Input:* "Find a dataset for kitchen hygiene and dirty surface detection."

- *Expected Output:* Queries both platforms and returns datasets related to food safety, dirty dishes, or clean/unclean surfaces with corresponding class names.
- **Eval Case 4:**
  - *Input:* "Find a dataset of purple-striped alien spaceships." (Edge Case: Non-existent data).
  - *Expected Output:* The agent searches, finds nothing, and explicitly states that no such dataset exists on these platforms, refusing to generate a fake URL.
- **Eval Case 5:**
  - *Input:* "Find a pedestrian detection dataset, but it must have over 10,000 images."
  - *Expected Output:* Returns datasets and successfully filters out any repository that has fewer than 10,000 images, explicitly stating the image count for the ones provided.

## 6. Risks and Guardrails

- **Risk (The "Swamp"):** The most critical risk is the agent hallucinating URLs. LLMs often invent plausible-looking links (e.g., [\[kaggle.com/datasets/user/pothole-yolo\]\(https://kaggle.com/datasets/user/pothole-yolo\)](https://kaggle.com/datasets/user/pothole-yolo)) that result in a 404 error, wasting the developer's time.
- **Guardrails:**
  1. The agent is strictly forbidden from generating a URL without first using the search tool to observe the actual search engine result page.
  2. The agent must verify the existence of the dataset name in the search results before constructing the link. It must never output a "guessed" URL.

## 7. Platform Choice and Justification Choice: Claude Project (or Claude Pro).

**Justification:** I chose Claude over an automation workflow like n8n or a scripted agent. While n8n is powerful for connecting APIs, building a reliable custom web scraper for dynamic, JavaScript-heavy sites like Kaggle and Roboflow would easily exceed the 10-hour build scope. Claude provides a native, highly capable web search tool combined with a massive context window. This allows the agent to ingest multiple search results, read the dataset descriptions, and reason about whether the classes actually match my engineering needs better than a simple API keyword filter could, making it the most efficient platform for this specific job.